Oraqle across readout datasets: main differences

Nov 6, 2026

Summary

The paper's patterns hold on other devices, but its durations do not. The best readout duration runs from 0.7 μs on the Lienhard chip to 15 μs on Zheng's fluxonium, and with QICK's readout the QEC spacetime optimum moves from 600 ns to 1.25 μs.

Paper's claim	What the other datasets show	Verdict
#1 Fidelity saturates by about 1 μs	Peaks at 1.3-1.8 μs on the fast devices, and at 8-15 μs on Zheng's devices and the bolometer	Shape holds, duration does not
#2 The first 0.5 μs holds most information	Later windows are far better on QICK, Cao and Hassani	Does not transfer
#3 Cutting to about 600 ns costs about 10% more error	CNN error is 4.2× on QICK and 1.8× on Cao	Principle holds, 600 ns does not
#4 Small models match much larger ones	KLiNQ, CNN and HERQULES match the FNN at each plateau, and HERQULES trails below it	Holds at the plateau, against the FNN only
#5 At 1 μs about 80% of errors are state changes	On QICK, 91% is overlap at 1 μs and 7% at 1.8 μs	Holds at each device's plateau
#9 600-800 ns is the QEC optimum	1.25 μs with QICK's readout, with the same sharp collapse below it	Shape holds, location moves

MCMit-T finished one run of 17, so "large model" here means the Baseline FNN. Takeaways #10-#14 were not tested.

The datasets

Five public datasets were compared with the Lienhard chip, and two of them are large enough to train every discriminator. All five read one qubit, so none can test crosstalk or multiplexing.

Dataset	Device	Shots used	Sample period	Labels	Used for
Lienhard et al. 2022 (not public)	5 transmons, multiplexed	1,600,000	2 ns	32 prepared states	Reference, plus a single-channel control
Di Guglielmo et al. 2025 (QICK)	1 transmon	1,000,000	3.25 ns	Prepared 0 or 1	Everything
Cao et al. 2025	1 transmon, pulse plus ring-down	207,179 after heralding	20 ns (inferred)	Prepared g or e; f kept apart	Everything but QEC
Hassani et al. 2023	Inductively shunted transmon	1,002 at each of 13 powers	50.8 ns	g or e by file	Readout sweeps
Zheng et al. 2026	1 transmon, 1 fluxonium	About 16,000 per duration	One integrated point per shot	g or e by run	True duration sweep
Gunyhó et al. 2024	Transmon read by a bolometer	2,000	64 ns	Pi pulse or none	Readout sweeps

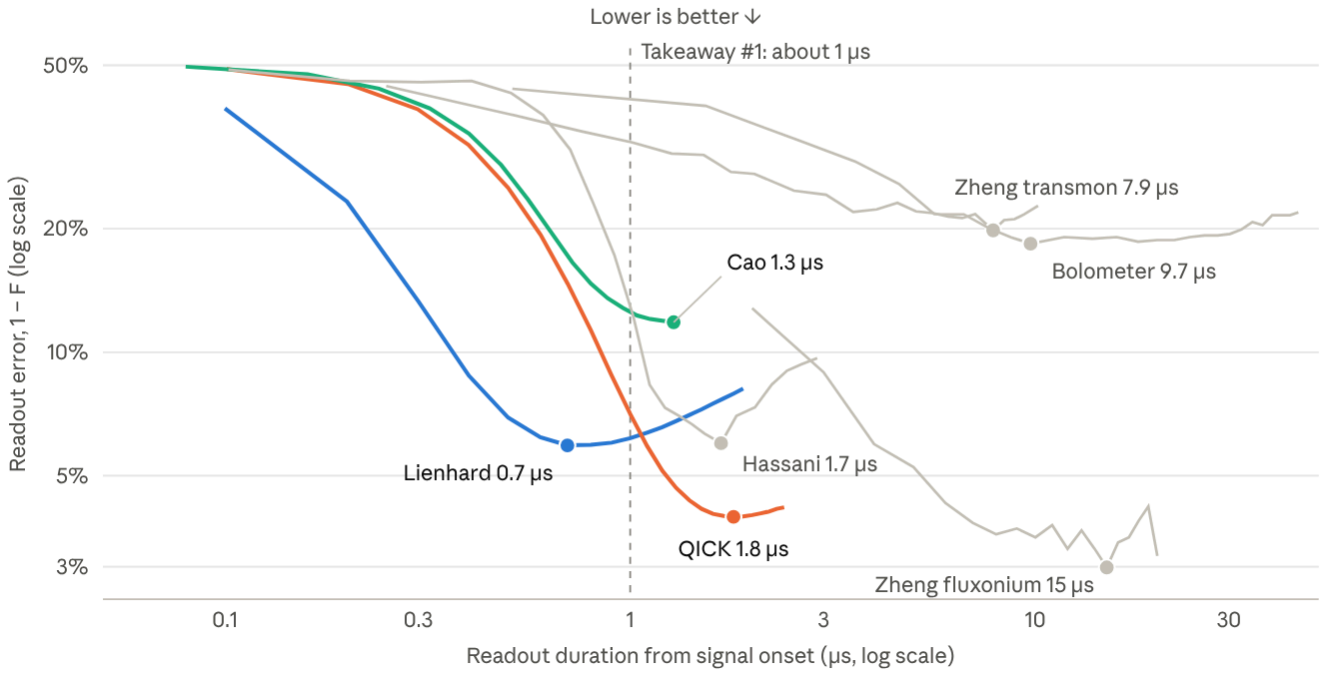
The analysis reproduces its sources: QICK's published 96%, and 0.63 against the bolometer paper's 0.618 once converted to its definition. On the Lienhard channels, one at a time, it lands a median 0.11 pp from Table 2 at 600 ns and above.

Where readout saturates

The best duration differs about 20-fold across devices, and only the Lienhard chip is near its minimum at 1 μs. QICK's error at 1 μs is still 1.8× its minimum.

The best readout duration runs from 0.7 µs to 15 µs across devices

Readout error of the linear baseline. Dots mark each minimum.

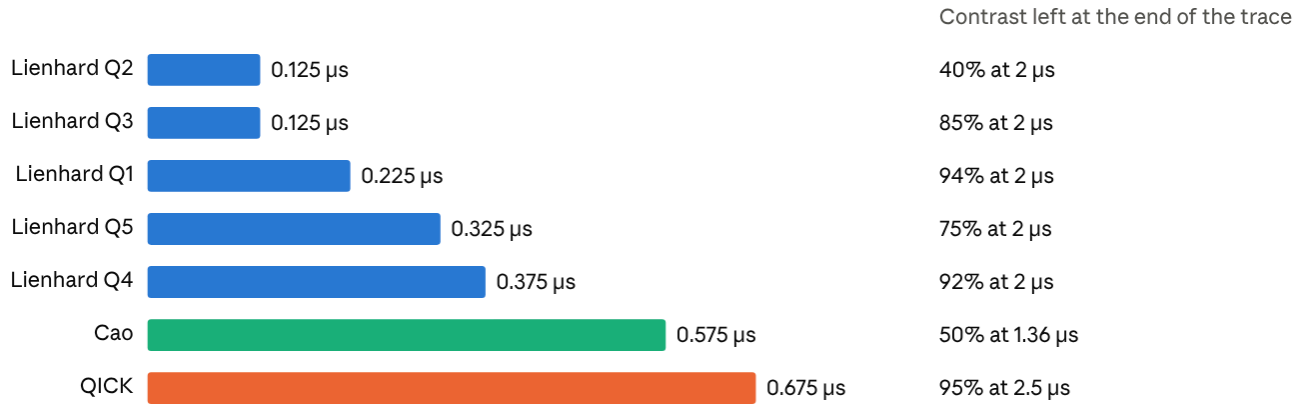


Oraqle extras/cross_dataset/results/linear_*.json. Lienhard: published sweep, geometric mean of Q1, Q3, Q4, Q5. Hassani at -16 dBm. Zheng: separate experiments at each duration.

Past the minimum the devices also differ: Lienhard's error grows by 2.2 pp between 0.7 and 1.9 µs, while QICK's grows by 0.2 pp between 1.8 and 2.4 µs. The likely cause is how fast state contrast builds and how long it lasts.

State contrast builds in 0.6-0.7 µs on QICK and Cao, 0.13-0.38 µs on Lienhard

Bars: time for the mean |1⟩-|0⟩ difference to reach 90% of its maximum, in 50 ns bins.



Oraqle extras/cross_dataset/exploration/output/risetime.log. Lienhard channels unfiltered, 32,000 shots.

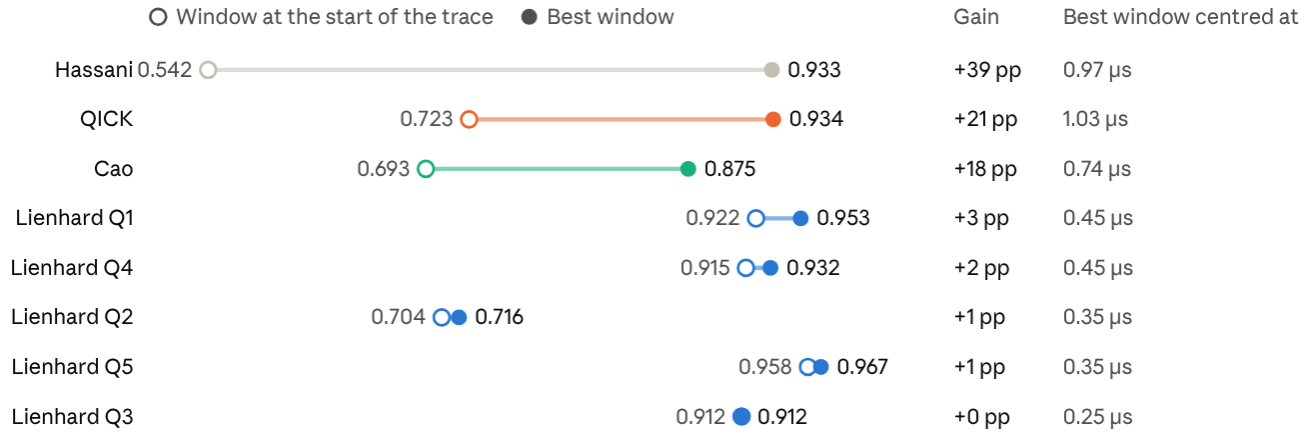
Slow build-up pushes the minimum later, and contrast that survives keeps the curve flat after it. This does not explain every qubit: Q2 builds as fast as any qubit yet peaks at 1.7 µs, because it is limited by signal-to-noise. Cao's contrast falls because its pulse ends after about 480 ns and the rest of its trace is ring-down.

Where the information sits

The first 500 ns holds most of the information only on the Lienhard chip. On the other devices the same window placed later scores 18-39 pp higher.

A later 500 ns window gains 18-39 pp elsewhere, at most 3 pp on Lienhard

Fidelity of the linear baseline integrated over one 500 ns window. Higher is better.



Oraqle extras/cross_dataset/results/linear_lp_qick.json, linear_lp_cao.json, linear_hassani.json (-16 dBm); Lienhard: published Baseline_Threshold_sliding_window_500ns.csv. All with the 10 MHz filter of the published windows, which has no effect at Hassani's sample rate.

A delayed window is not the answer in itself, because latency counts from the start of the pulse. At the same end time, integrating from the start still beats the 500 ns window on QICK: 0.95 against 0.93 at 1.3 µs.

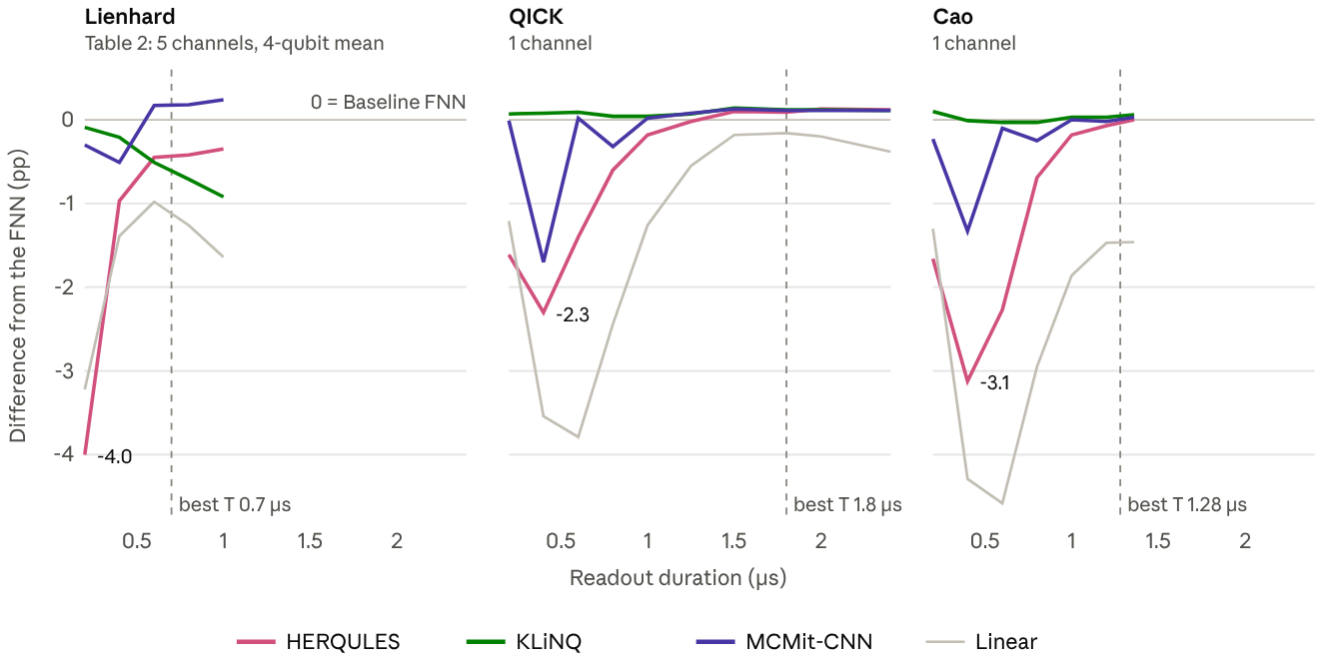
Dropping only the first samples does help. Skipping the first 200 ns lifts the linear baseline at 600 ns from 0.805 to 0.833 on QICK and from 0.775 to 0.802 on Cao. On the Lienhard qubits the best skip gains at most 0.6 pp. Learned models do this on their own: the CNN reaches 0.844 at 600 ns on QICK.

Discriminators

Small models match the Baseline FNN once a device reaches its plateau, on every dataset. Below the plateau they split: KLiNQ keeps up on QICK and Cao, and HERQULES falls furthest behind.

Small models reach the FNN at each plateau; HERQULES trails it by up to 4 pp below

Fidelity difference from the Baseline FNN, in percentage points. Higher is better.



Oraqle extras/cross_dataset/results/summary.json; Lienhard from results/discriminators/master_fidelity.csv. Dashed line: each device's best duration from the linear baseline. Paired bootstrap intervals are in the Oraqle README.

KLiNQ's -0.9 pp on Lienhard comes from its raw multiplexed input; on a demodulated channel it ties the others.

What a learned model adds differs by device. On QICK the best model's gain over the linear baseline falls from 3.9 pp at 600 ns to 0.3 pp at 1.8 µs, because contrast is steady once built. On Lienhard it grows from 1.2 pp at 600 ns to 1.9 pp at 1 µs, likely because late samples carry decay that a fixed integrator cannot discount.

Model (parameters, one qubit)	Lienhard 1 µs	QICK 1 µs	QICK 1.8 µs	Cao 1 µs	Cao 1.36 µs
Linear (5)	0.938	0.929	0.960	0.872	0.882
QubiCML (65)	0.939	0.929	0.960	0.877	0.886
HERQULES (292)	0.951	0.940	0.962	0.889	0.896
KLiNQ (705)	0.946	0.942	0.963	0.891	0.897
MCMit-CNN (2,609)	0.957	0.942	0.962	0.891	0.897
Baseline FNN (238,856)	0.955	0.942	0.961	0.891	0.896

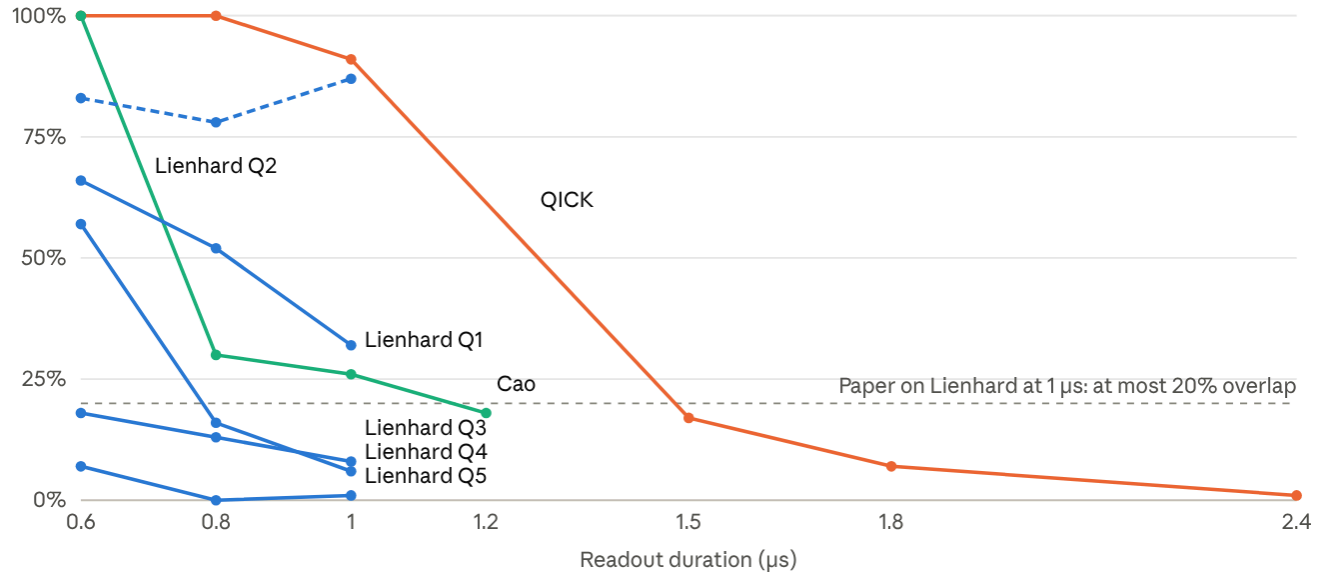
The Lienhard column is Table 2's mean of Q1, Q3, Q4 and Q5 on five channels. Levels should not be compared across devices, because Cao's are limited by its labels.

What the remaining errors are

Overlap shrinks with duration and is a minority by each device's plateau, so the paper's split holds there, not at a fixed 1 µs. Lienhard Q2 is the exception, because it is SNR-bound. On QICK the crossover lies between 1 and 1.5 µs.

At 1 µs overlap is 91% of QICK error, against 1-32% on Lienhard Q1, Q3, Q4, Q5

Gaussian overlap as a share of the CNN error, capped at 100%. The rest is state change, preparation or leakage.



Oraqle extras/cross_dataset/results/error_modes.json, single-channel CNN on every dataset. Dashed Lienhard Q2 is the paper's SNR-bound qubit.

This estimate is a proxy for the paper's HMM pipeline: a Gaussian fitted to the far side of each state's score distribution. Pooled over Lienhard's Q1, Q3 and Q5 it gives 12%, against the paper's bound of 17%, and it is rougher per qubit.

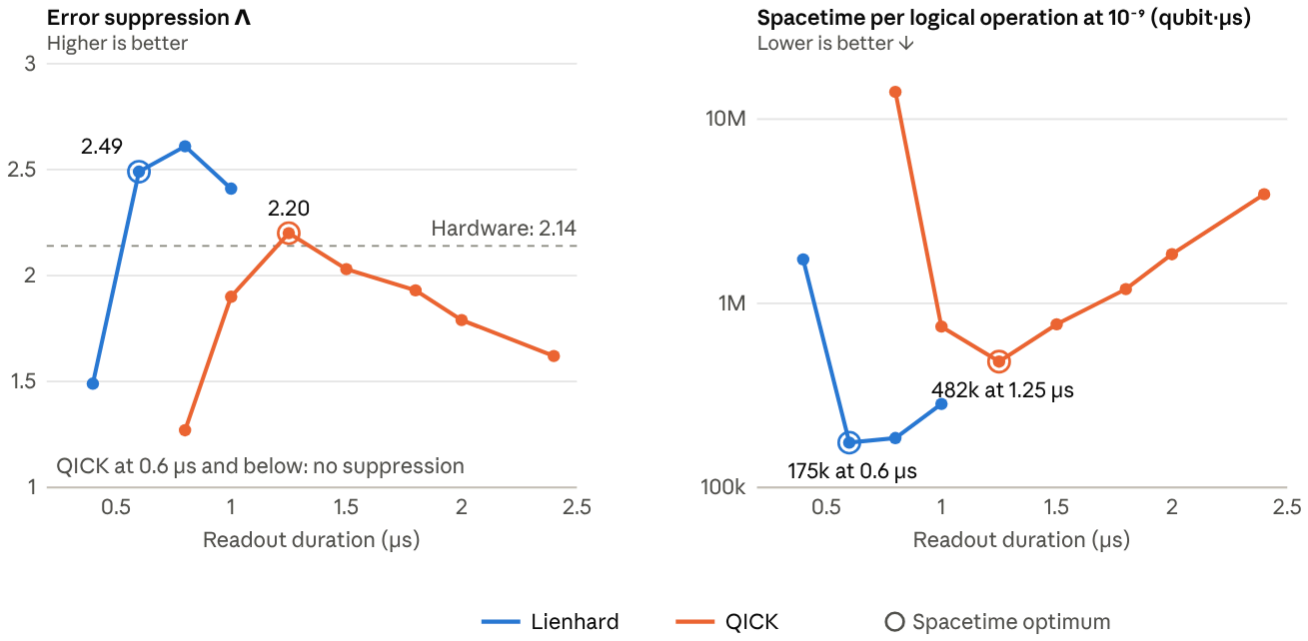
Beyond overlap, P(0|1) exceeds P(1|0) on every device: 5.0% against 2.5% on QICK at 1.8 µs, 15.2% against 5.4% on Cao at 1.36 µs, and 4.6% against 1.4% on Lienhard Q1 at 1 µs.

QEC impact

With the same gates, QICK's readout moves the spacetime optimum from 600 ns to 1.25 µs and costs 2.8× as much at that optimum. The shape of Takeaway #9 survives; its location does not.

QICK readout moves the QEC optimum from 0.6 to 1.25 µs and costs 2.8× the spacetime

MCMit-CNN readout through the paper circuit and ibm_boston gates. Only the readout changes.



Lienhard: Oraqle results/system/ler_disc_hi.jsonl (published). QICK: extras/cross_dataset/results/ler_qick.jsonl. Both fitted by system/analyze.py; d from 5 to 15.

The optimum saves about the same against 1 µs: 0.65× on QICK and 0.62× on Lienhard. Below it the cost collapses again, reaching 29× at 800 ns on QICK. QICK's optimum has the lower readout error (4.3% against 5.1%), but its rounds are 1.7× longer, which adds idle error and lowers Λ. The dashed line is the Λ of 2.14 measured on hardware, as §7 cites it (Acharya et al. 2025).

The two optima may share a device-independent form: each sits where readout error is about 1.2× its floor (1.17× on QICK, 1.18× on Lienhard at 600 ns). This is a hypothesis from two devices on a coarse grid, not a result.

Limits

The QICK and Cao comparisons are the firmest, because every model ran on the same large test sets. The QEC result carries the largest open question.

- **QEC circuit.** The spacetime optimum appears only in lattice-sim's lattice-surgery circuit, which all of §7 uses. There, 600 ns costs 0.62× of 1 µs on the Lienhard chip. Three memory circuits give 0.94-1.30×, and the cause is not established. QICK's optimum comes from the same circuit.
- **QEC gates.** The QICK result pairs its readout with the paper's `ibm_boston` gates. It is not a prediction for the QICK device, whose T1 is not in its deposit.
- **MCMit-T.** One run of 17 finished, so the largest model is untested on the new data.

- **Channels.** The Lienhard numbers use five channels, and QICK and Cao use one. The single-channel control shows this matters for Q3: the CNN drops from 0.942 to 0.927 at 1 μs.
- **Training data.** The FNN draws from a 320,000-shot subsample, as in the paper, while the CNN and KLiNQ use up to 810,000. So its small deficit at QICK's plateau may come from data, not capacity. It disappears on the smaller Cao set.
- **Labels.** Cao's fidelity sits near 0.90 mainly because of its labels. Heralding errors and decay leave about 15% of prep_e shots outside the e blob.
- **Time axes.** Cao's 20 ns sample period is inferred, and the bolometer's T = 0 comes from its state contrast, within about 1 μs.
- **Small sets.** Hassani has 300 test shots per power, so about 1.4 pp per point, and -16 dBm is the best of 13 powers. The bolometer has 2,000 shots.

The code, per-run results and logs are in `extras/cross_dataset/` in the Oraqle repo.